

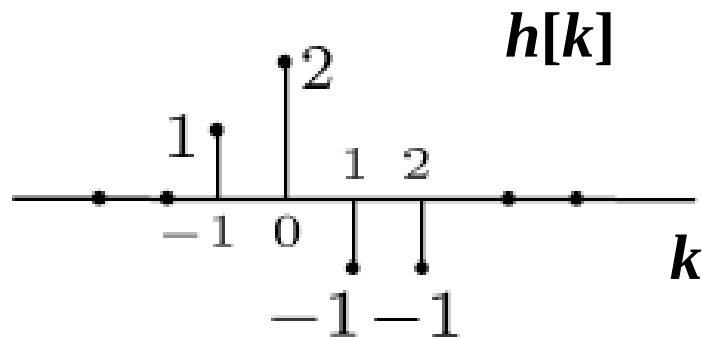
Assignments (Week 4)

- **2.4**
- **2.6**
- **2.19**
- **2.21 (c) (d)**

Tutorial Problems (Week 5)

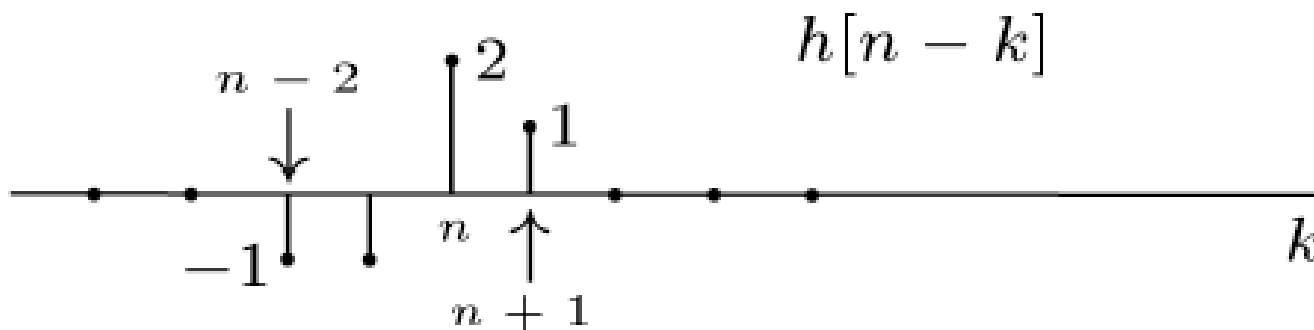
- **Basic Problems with Answers 2.3, 2.7, 2.13**
- **Basic Problems 2.24, 2.26**

- Time-shift and flip



What is the plot for $h[n-k]$??
 n is a constant

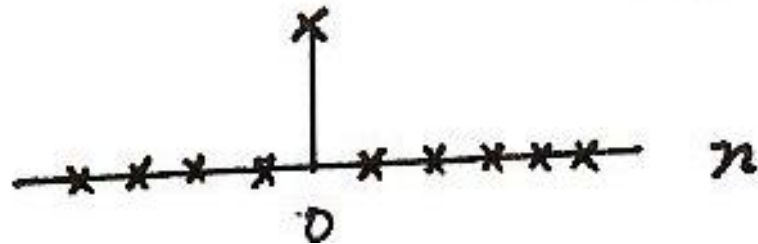
$$h[k] \xrightarrow{\text{Flip}} h[-k] \xrightarrow{\text{Slide}} h[n-k]$$



- **Unit impulse function (unit sample function)**

Discrete-time

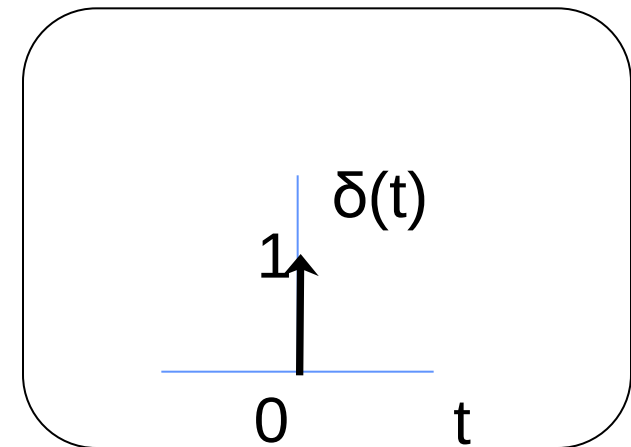
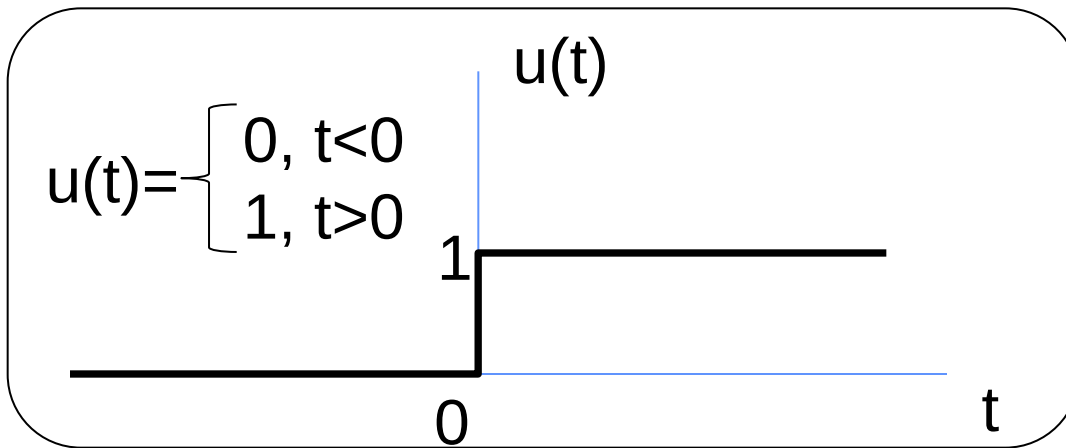
$$\delta[n] = \begin{cases} 0, & n \neq 0 \\ 1, & n = 0 \end{cases}$$



- We can use unit impulse function to represent any other different signals, or it is a building function (or basic signal).

Unit Impulse and Unit Step Functions

Continuous-time



Relation between unit impulse and unit step functions

– First Derivative

$$\delta(t) = \frac{du(t)}{dt}$$

– Running Integral

$$u(t) = \int_{-\infty}^t \delta(\tau) d\tau$$

Unit Impulse and Unit Step Functions (Cont.)

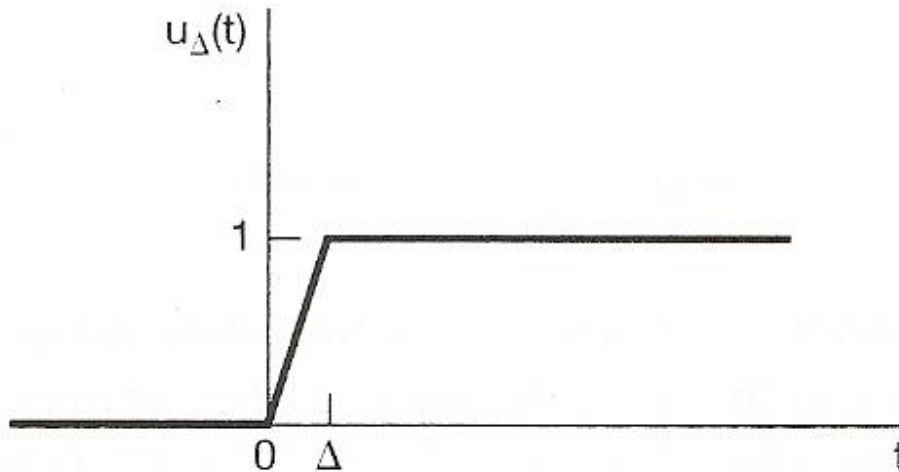


Figure 1.33 Continuous approximation to the unit step, $u_{\Delta}(t)$.

$$\delta_{\Delta}(t) = \frac{du_{\Delta}(t)}{dt}$$

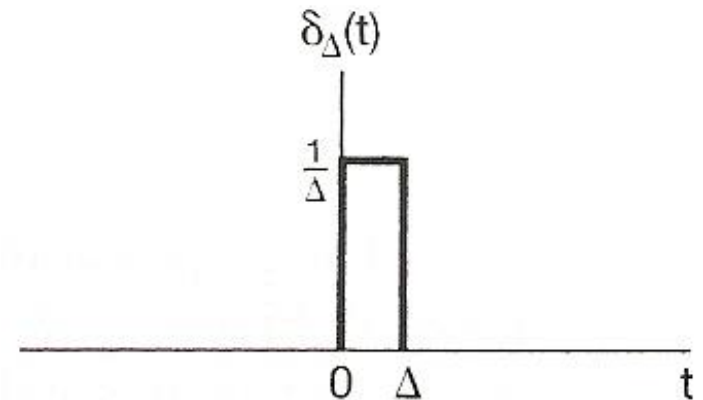


Figure 1.34 Derivative of $u_{\Delta}(t)$.

System properties:

1. With memory or memoryless

$$y(n) = f(x(n))$$

2. Invertible

for a system $x \rightarrow y$, if $x_1 \neq x_2$, then $y_1 \neq y_2$

3. Causal

... up to that time n ...

4. Stable

either prove the system is stable, or find a specific counterexample

5. Time-invariant

$$\begin{array}{ll} \text{If} & x[n] \rightarrow y[n] \\ \text{then} & x[n - n_0] \rightarrow y[n - n_0] . \end{array}$$

6. Linear

A (CT) system is linear if it has the **superposition property**:

$$\text{If} \quad x_1(t) \rightarrow y_1(t) \quad \text{and} \quad x_2(t) \rightarrow y_2(t)$$

$$\text{then} \quad ax_1(t) + bx_2(t) \rightarrow ay_1(t) + by_2(t)$$

Chapter 2

Linear Time-invariant (LTI) Systems

Exploiting Superposition and Time-Invariance

If we have $x_k[n] \rightarrow y_k[n]$, then

$$x[n] = \sum_k a_k x_k[n] \xrightarrow{\text{Linear System}} y[n] = \sum_k a_k y_k[n]$$

Question: Are there sets of “basic” signals $x_k[n]$ such that

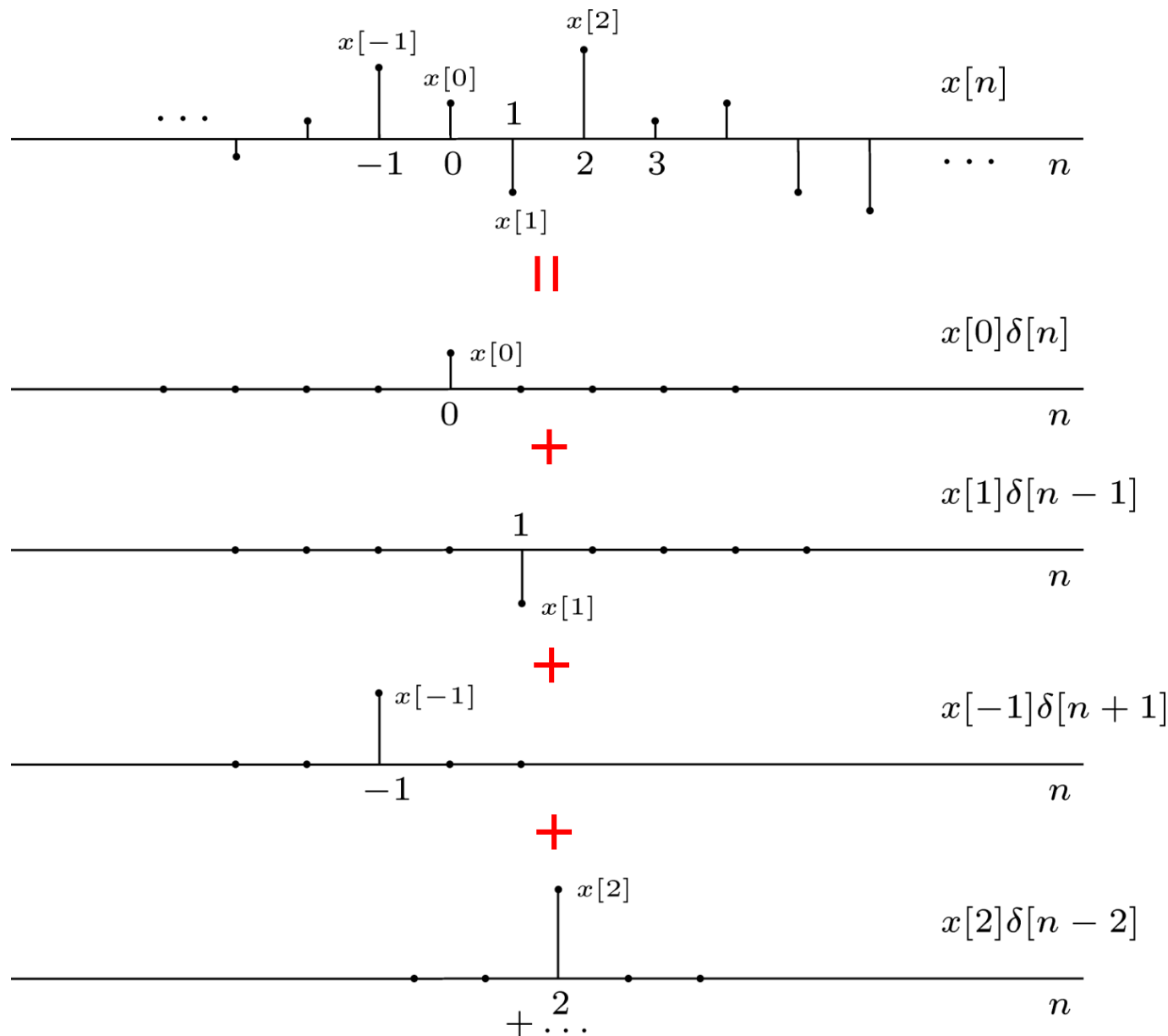
- a) We can represent *any* signals as linear combinations of these building block signals.
- b) The response of LTI Systems to these basic signals are both *simple* and *insightful*.

Fact: For LTI Systems (CT or DT) there are two natural choices for these building blocks

Focus for now: DT Shifted unit samples $\delta[n-n_o]$

Next time: CT Shifted unit impulses $\delta(t-t_o)$

Representation of DT Signals Using Unit Samples ¹⁰



That is ...

$$x[n] = \dots + x[-2]\delta[n+2] + x[-1]\delta[n+1] + x[0]\delta[n] + x[1]\delta[n-1] + \dots$$

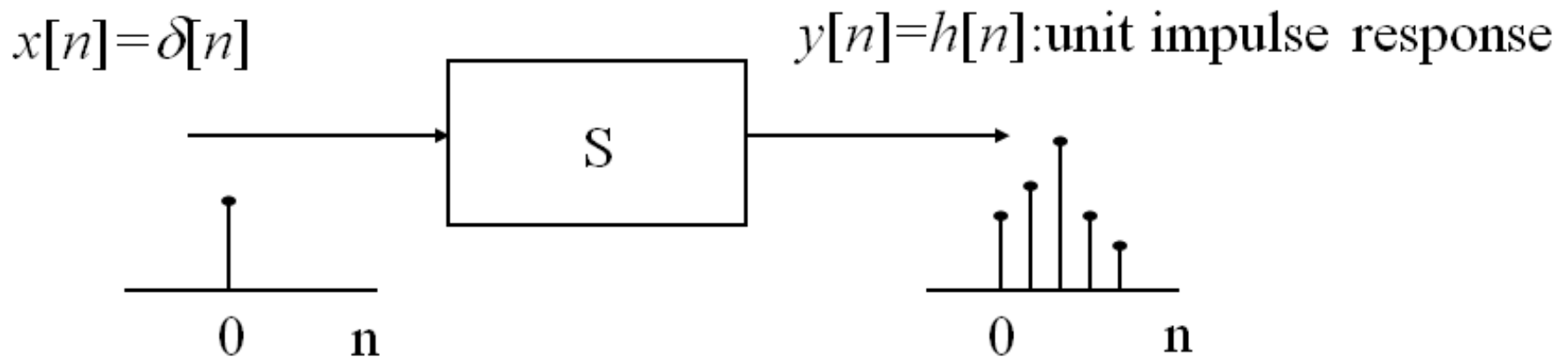
Important to note the “-” sign

$$x[n] = \sum_{k=-\infty}^{+\infty} \underbrace{x[k]}_{\text{Coefficients}} \underbrace{\delta[n-k]}_{\text{Basic Signals}}$$

- **Sifting property** of the unit impulse: looked at the index k , $\delta[n-k]$ is nonzero only at $k = n$, which “sifts” the value $x[n]$ out of the function $x[k]$.

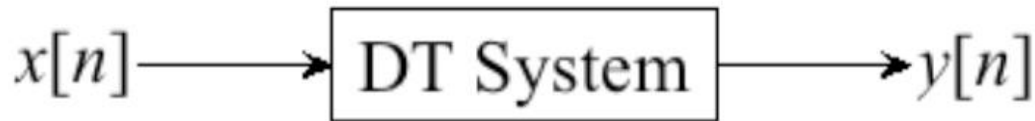
Unit Impulse Response (Unit Sample Response)

- Define the output for an **unit impulse input** as the **unit impulse response**



Example: $y[n] = x[n] + 2x[n-1] + 4x[n-2]$
 What is unit impulse response?

Response of DT LTI Systems



- Now suppose the system is **LTI**, and define the *unit impulse response* $h[n]$:

$$\delta[n] \longrightarrow h[n]$$



From **T**ime-**I**nvariance:

$$\delta[n - k] \longrightarrow h[n - k]$$

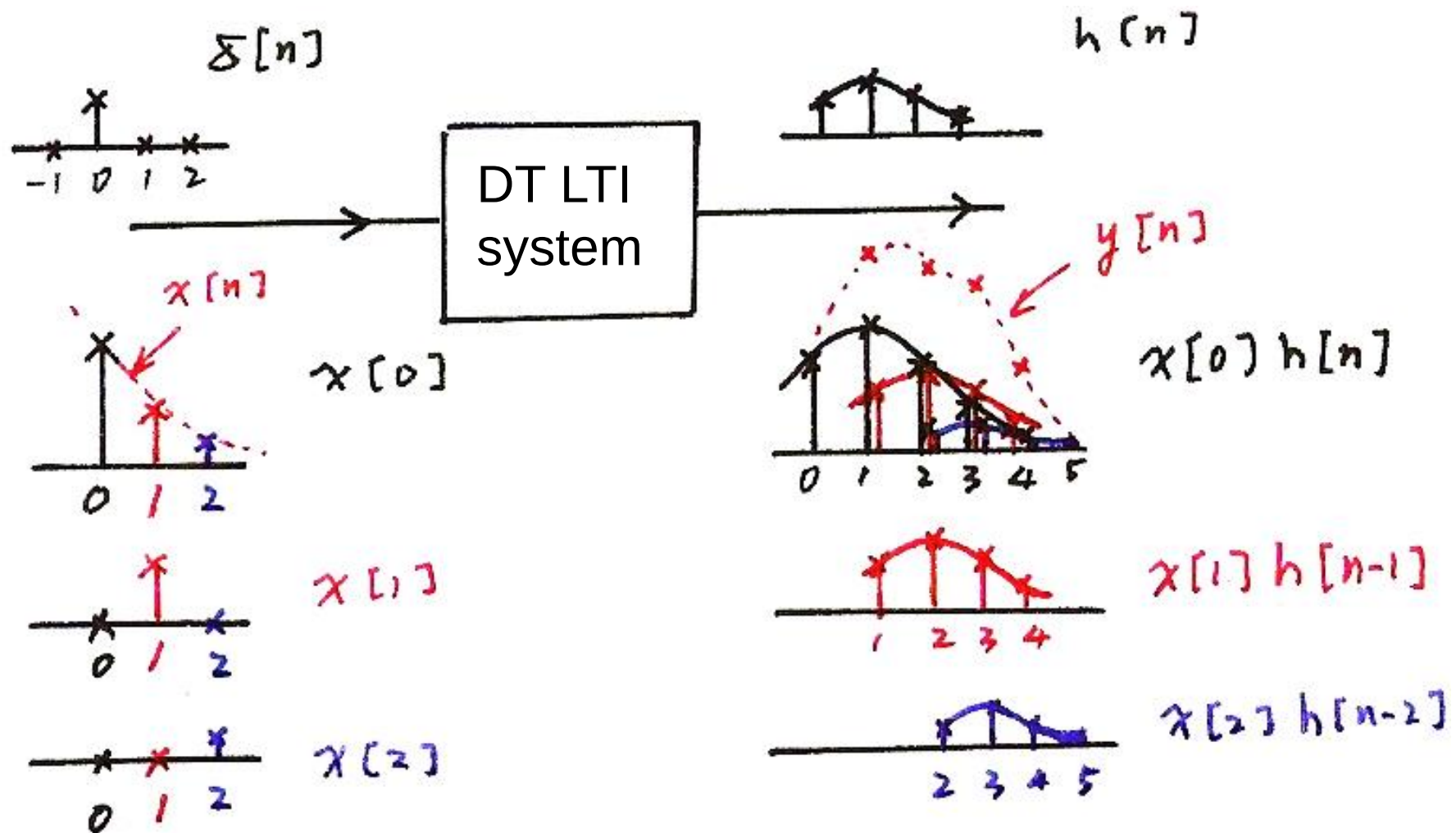
From **L**inearity:

$$x[n] = \sum_{k=-\infty}^{+\infty} x[k] \delta[n - k] \longrightarrow y[n] = \underbrace{\sum_{k=-\infty}^{+\infty} x[k] h[n - k]}_{\text{convolution sum}} = x[n] * h[n]$$

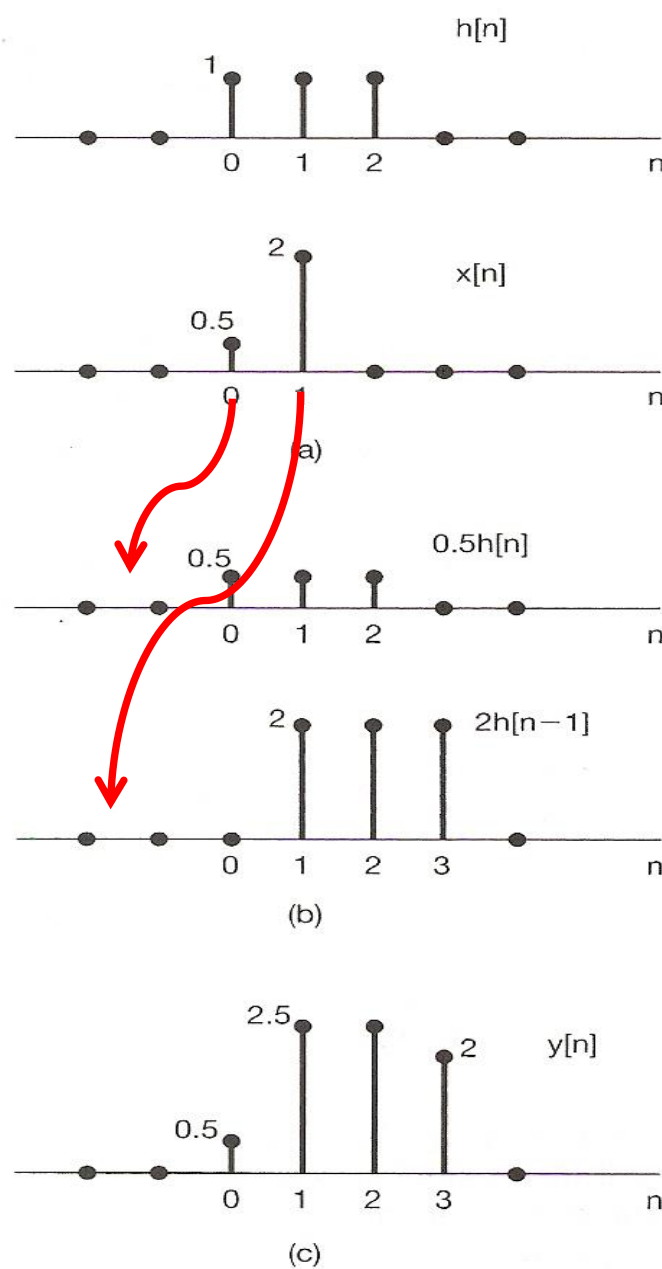
The output for an arbitrary input signal is the superposition of a series of “shifted, scaled unit impulse response”

Example

Input/Output Relation



Example



$$y[n] = 0.5h[n] + 2h[n-1]$$

Figure 2.3 (a) The impulse response $h[n]$ of an LTI system and an input $x[n]$ to the system; (b) the responses or “echoes,” $0.5h[n]$ and $2h[n-1]$, to the nonzero values of the input, namely, $x[0] = 0.5$ and $x[1] = 2$; (c) the overall response $y[n]$, which is the sum of the echoes in (b).

Hence a Very Important Property of LTI Systems:

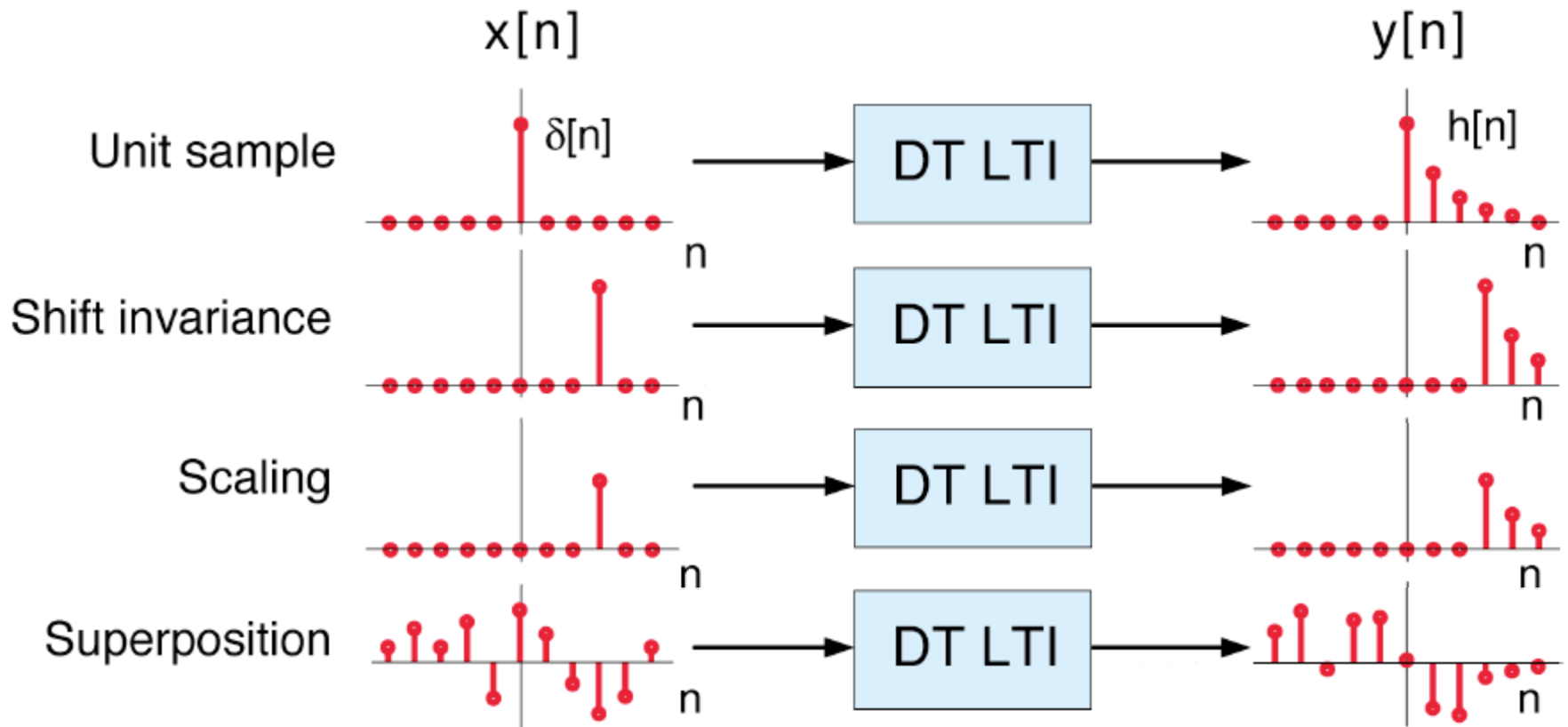
The output of *any* DT LTI System is a convolution of the input signal with the unit-sample response, *i.e.*

$$\begin{aligned}\text{Any DT LTI} &\longleftrightarrow y[n] = x[n] * h[n] \\ &= \sum_{k=-\infty}^{+\infty} x[k] h[n - k]\end{aligned}$$

As a result, any DT LTI Systems are *completely characterized* by its unit sample response

Graphic View of the Convolution Sum

Response of DT LTI systems



$$x[n] = \sum_{k=-\infty}^{+\infty} x[k] \delta[n - k] \xrightarrow{\text{DT LTI}} y[n] = \underbrace{\sum_{k=-\infty}^{+\infty} x[k] h[n - k]}_{\text{convolution sum}}$$

- A different way to visualize the convolution sum
 - looked at on the index k

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

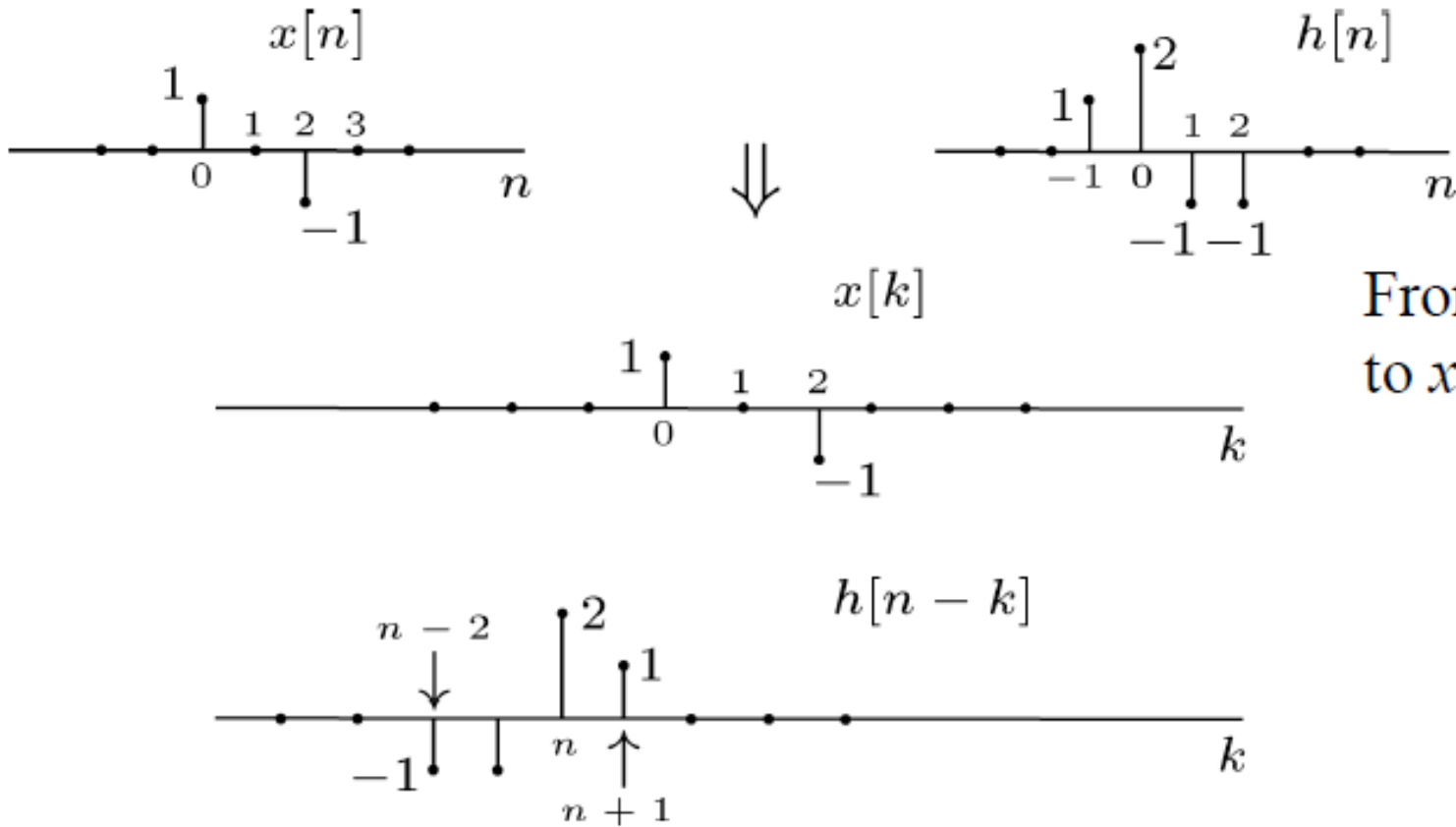
Contribution to the
output signal at time n

input signal

flipped version of $h[k]$ located at $k = n$

- on the dummy index k , $h[k]$ is **flipped** over and **shifted** to $k=n$, **weighted** by $x[k]$, and **summed** to produce an output sample $y[n]$ at time n

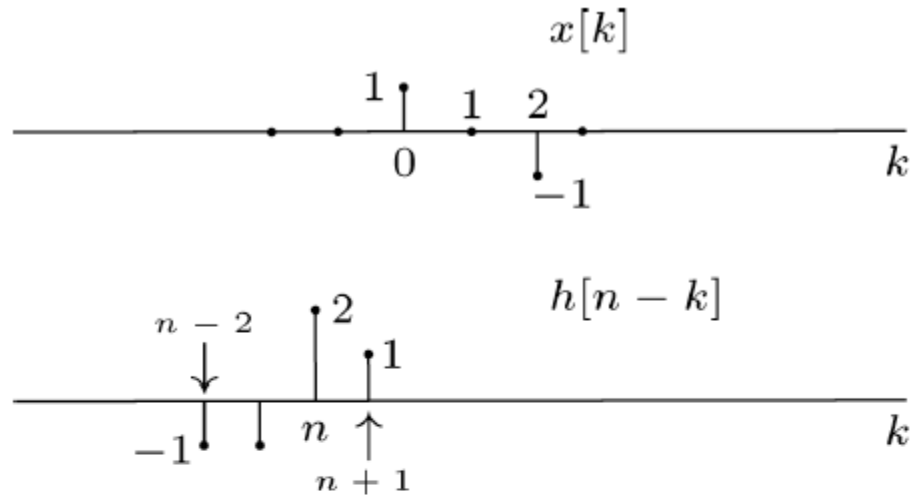
Example: Flip $h[n]$



From $x[n]$ and $h[n]$
to $x[k]$ and $h[n-k]$

Calculating Successive Values: **Shift,** **Multiply, Sum**

$$y[n] = \sum_{k=-\infty}^{+\infty} x[k]h[n-k]$$



$$y[n] = 0 \quad \text{for } n <$$

$$y[-1] =$$

$$y[0] =$$

$$y[1] =$$

$$y[2] =$$

$$y[3] =$$

$$y[4] =$$

$$y[n] = 0 \quad \text{for } n >$$

Convolution operation procedure:

$$h[k] \xrightarrow{\text{Flip}} h[-k] \xrightarrow{\text{Slide}} h[n-k] \xrightarrow{\text{Multiply}} x[k]h[n-k]$$

F S M S

$$\xrightarrow{\text{Sum}} \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$



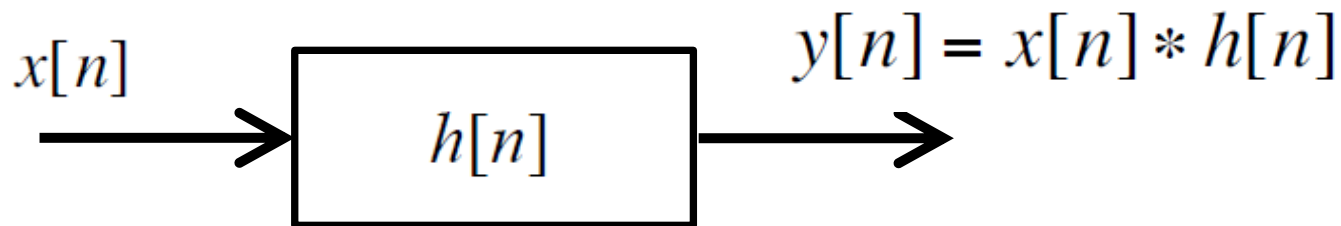
$$y[n]$$

$$\text{Any DT LTI} \iff y[n] = x[n] * h[n]$$
$$= \sum_{k=-\infty}^{+\infty} x[k] h[n - k]$$

$x[n]$: input

$y[n]$: output

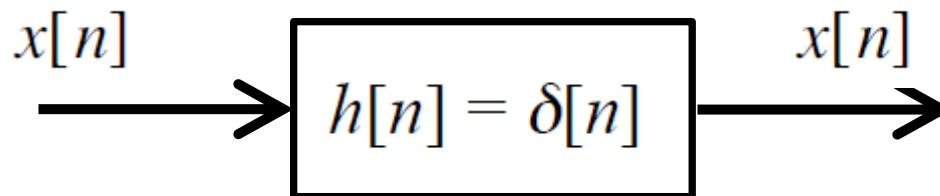
$h[n]$: impulse response of the system



Examples of Convolution and DT LTI Systems

Ex. #1: $h[n] = \delta[n]$

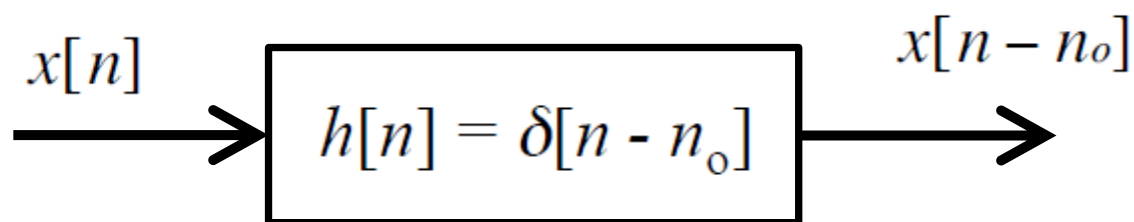
$$\begin{aligned} y[n] &= x[n] * \delta[n] = \sum_{k=-\infty}^{\infty} x[k] \delta[n-k] \\ &= x[n] \quad \text{— An Identity system} \end{aligned}$$



- sifting property, i.e., convolution sum (or integral) with a unit impulse function gives the original signal

Ex. #2: $h[n] = \delta[n - n_o]$

$$\begin{aligned} y[n] &= x[n] * \delta[n - n_o] = \sum_{k=-\infty}^{\infty} x[k] \delta[n - n_o - k] \\ &= x[n - n_o] \quad \text{— A Shift} \end{aligned}$$



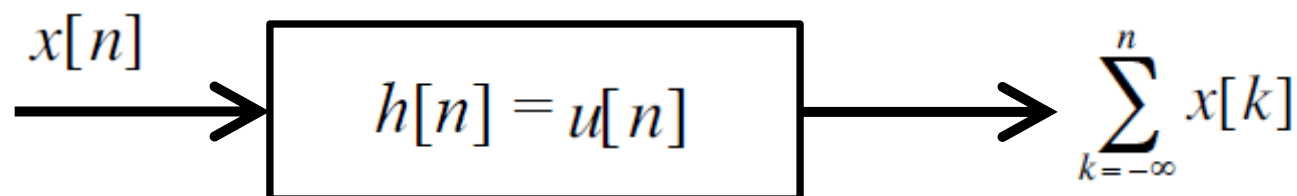
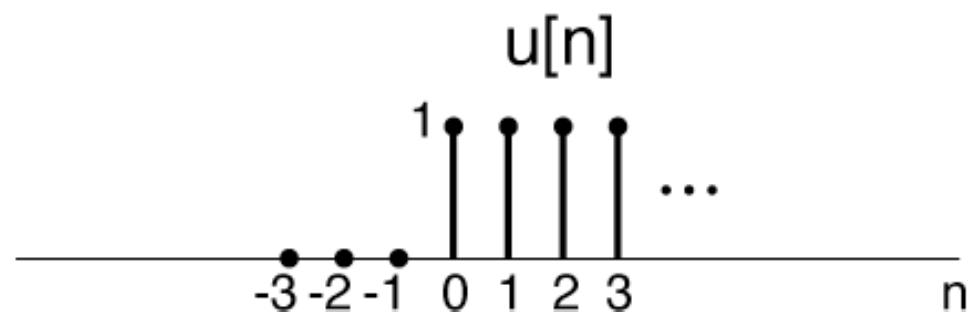
Ex. #3 $y[n] = \sum_{k=-\infty}^n x[k]$ – An accumulator

Unit Sample response

$$h[n] = \sum_{k=-\infty}^n \delta[k] = u[n]$$

↓

$$x[n] * u[n] = \sum_{k=-\infty}^n x[k]$$



Ex. #4 (Example 2.3)

$$0 < \alpha < 1$$

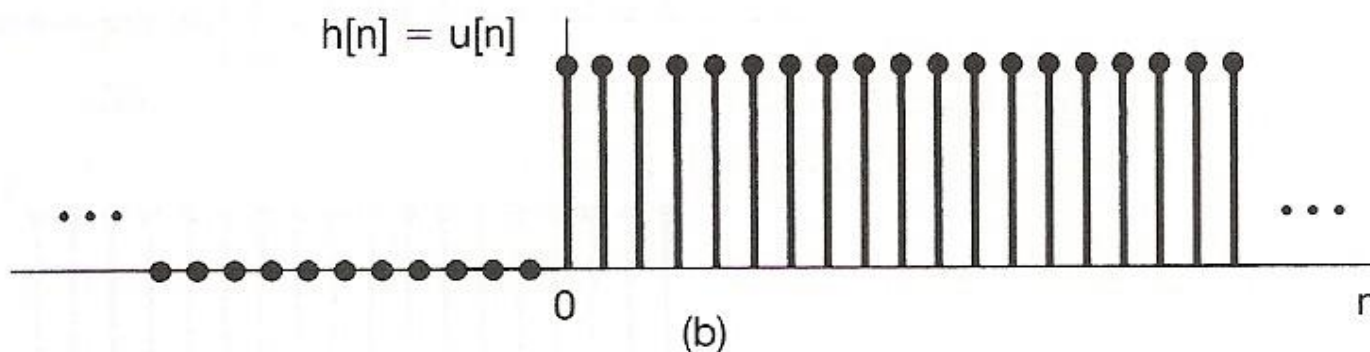
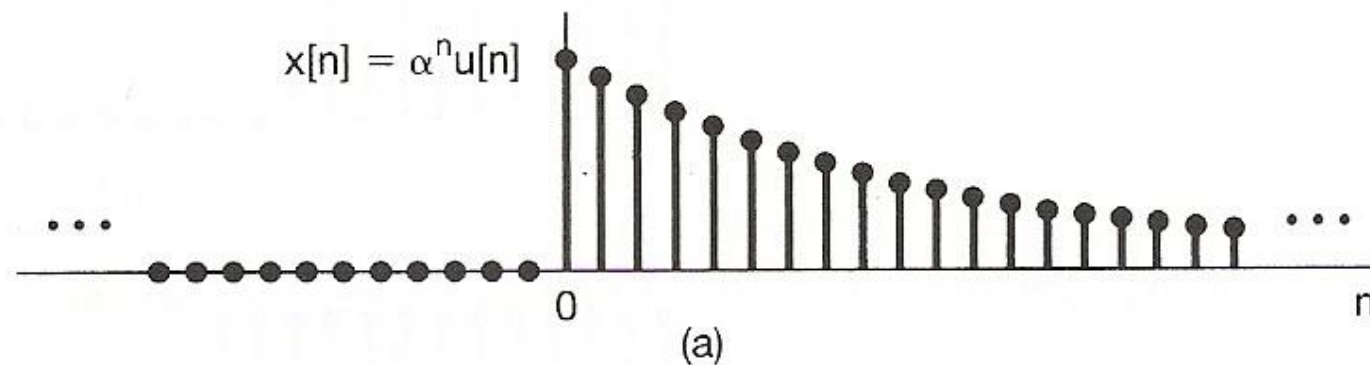


Figure 2.5 The signals $x[n]$ and $h[n]$ in Example 2.3.

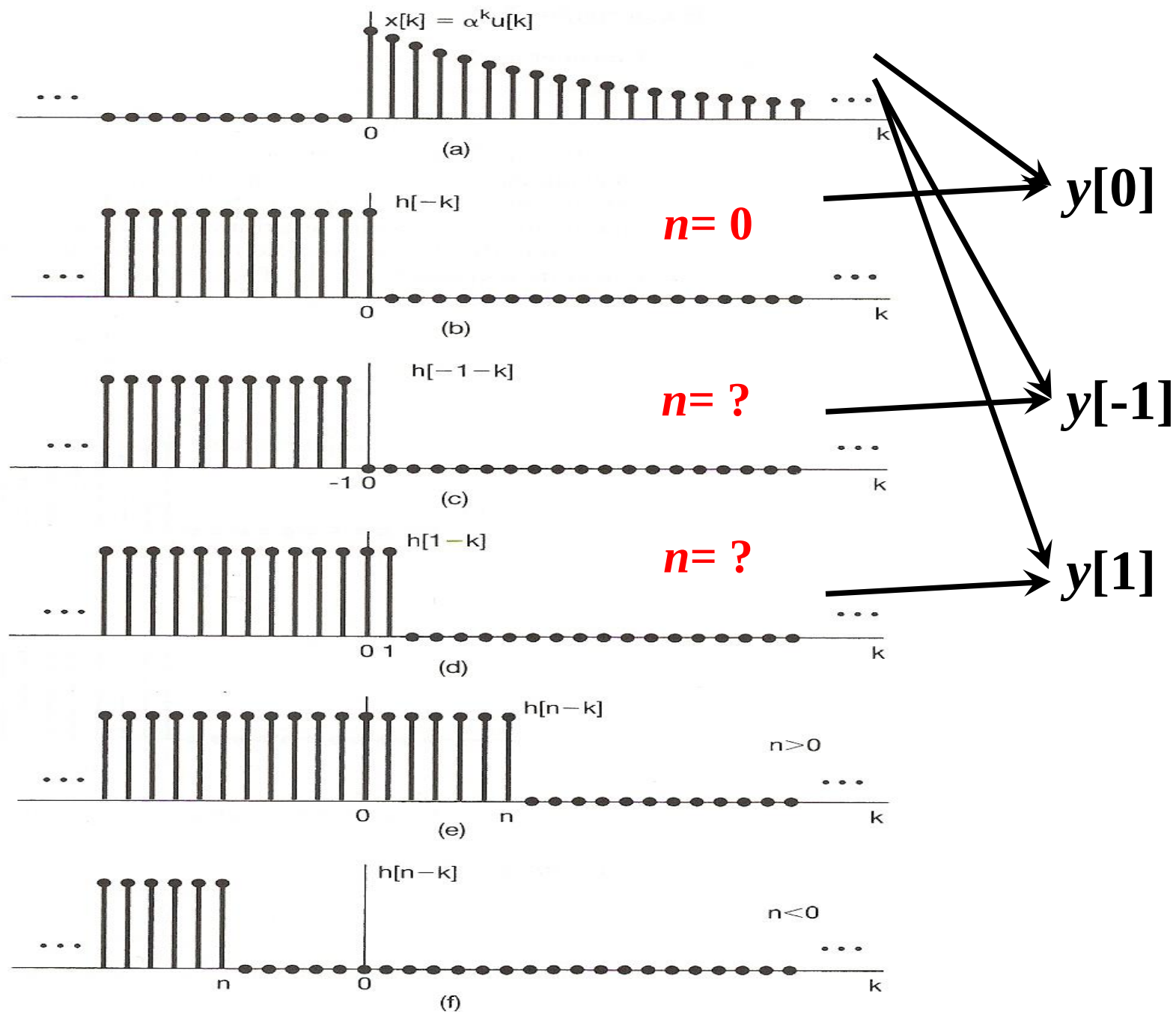


Figure 2.6 Graphical interpretation of the calculation of the convolution sum for Example 2.3.

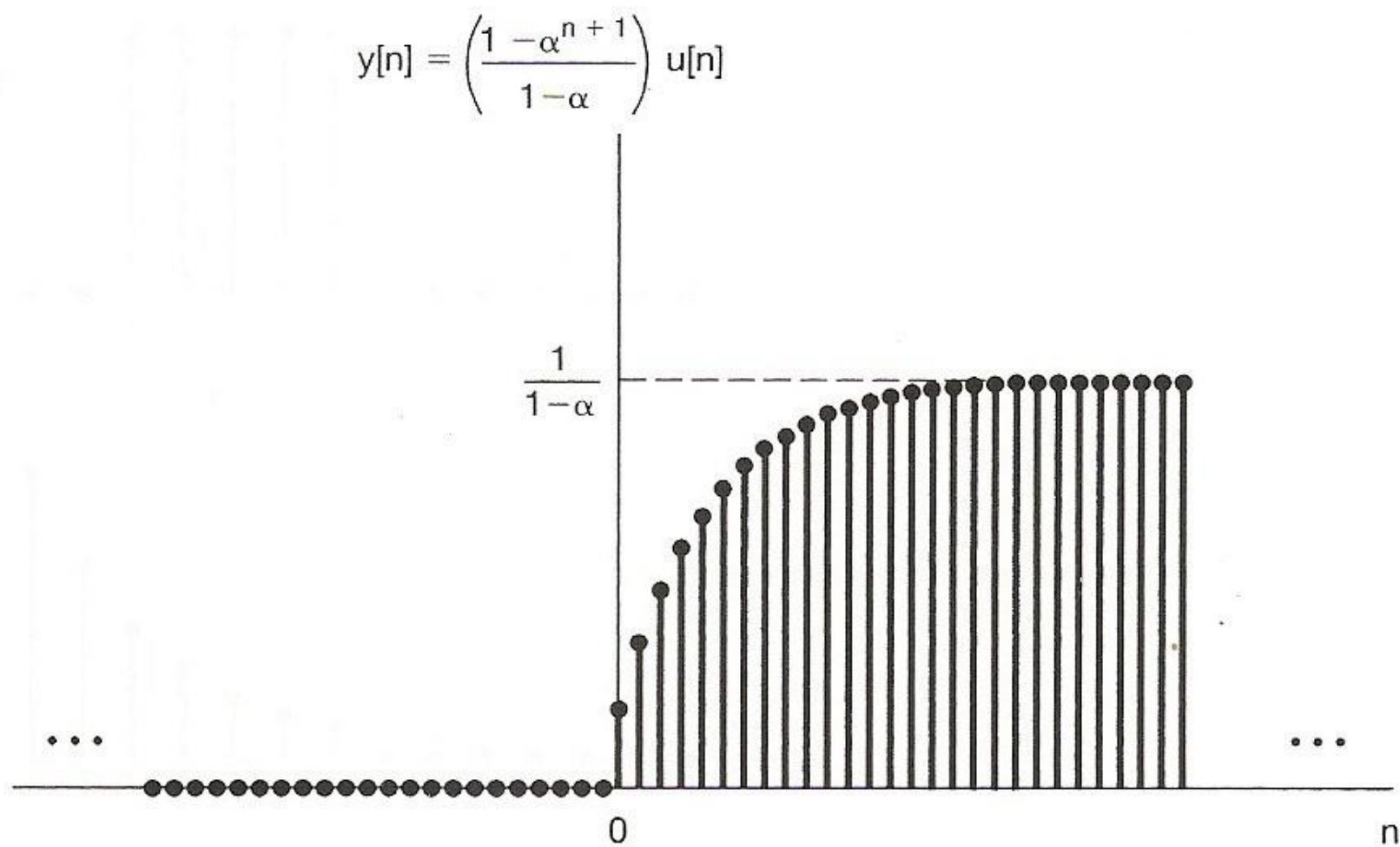


Figure 2.7 Output for Example 2.3.

● Characteristics of an LTI system are completely determined by its impulse response.

● *What if the system is nonlinear?*

Consider a discrete-time system with unit impulse response

$$h[n] = \begin{cases} 1 & n = 0, 1 \\ 0 & \text{otherwise} \end{cases}$$

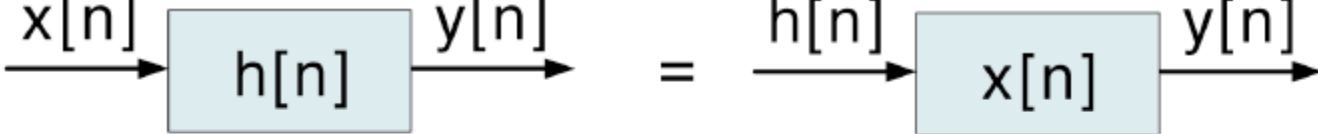
If the system is LTI, the input/output relationship is

$$y[n] = x[n] + x[n - 1].$$

On the other hand, there are *many* nonlinear systems with the same response to the input $\delta[n]$.

$$\begin{aligned} y[n] &= (x[n] + x[n - 1])^2, \\ y[n] &= \max(x[n], x[n - 1]). \end{aligned}$$

The Commutative Property of Convolution

$$y[n] = x[n] * h[n] = h[n] * x[n]$$


Example: Step response $s[n]$ of an LTI system **input: unit step function**

$$s[n] = u[n] * h[n] = h[n] * u[n]$$

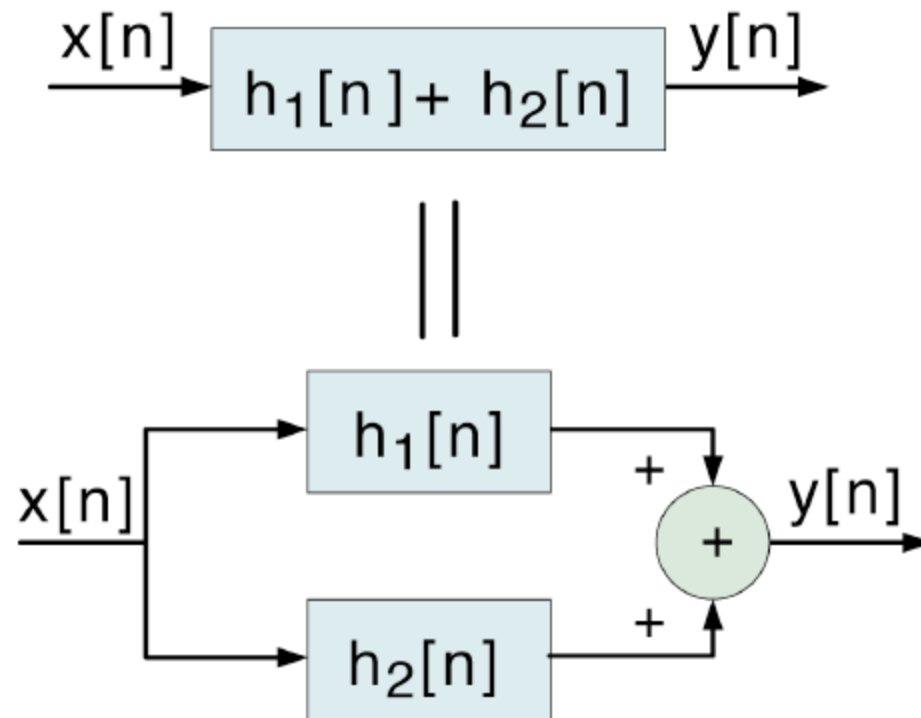
$\uparrow$ $\uparrow$
 “Input” Unit Sample response
 of accumulator

$$s[n] = \sum_{k=-\infty}^n h[k]$$

The Distributive Property of Convolution

$$x[n] * \{h_1[n] + h_2[n]\} = x[n] * h_1[n] + x[n] * h_2[n]$$

Interpretation



The Associative Property of Convolution

$$x[n] * (h_1[n] * h_2[n]) = (x[n] * h_1[n]) * h_2[n]$$

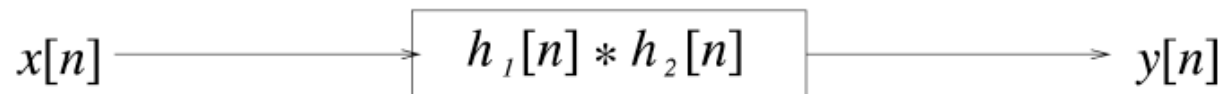
(Commutativity) ||

$$x[n] * (h_2[n] * h_1[n]) = (x[n] * h_2[n]) * h_1[n]$$

Implication (Very special to LTI Systems)



||



||



Properties of Convolution

Combining the Commutative property,

$$y[n] = x[n] * h[n] = h[n] * x[n]$$

Distributive property,

$$x[n] * \{h_1[n] + h_2[n]\} = x[n] * h_1[n] + x[n] * h_2[n]$$

and Associative property,

$$x[n] * (h_1[n] * h_2[n]) = (x[n] * h_1[n]) * h_2[n]$$

symbolically, we can treat “*” as a “×”. Easy, piece of cake!

The hard part is the actual calculation of the convolution.

Flip → Slide → Multiply → Sum.

Soon we will develop a clever way (*transformation*) to perform “×” instead of “*” operation.

Some Useful Properties of LTI Systems

1) Causality $\Leftrightarrow h[n] = 0$ for all $n < 0$

2) Stability $\Leftrightarrow \sum_{k=-\infty}^{+\infty} |h[k]| < \infty$

BIBO — Bounded Input $\Rightarrow$ Bounded Output

→ Sufficient condition: For $|x[n]| \leq x_{\max} < \infty$,

$$|y[n]| = \left| \sum_{k=-\infty}^{\infty} x[k]h[n-k] \right| \leq x_{\max} \left| \sum_{k=-\infty}^{\infty} h[n-k] \right| < \infty.$$

→ Necessary condition: If $\sum_{k=-\infty}^{\infty} |h[k]| = \infty$

Let $x[n] = h^*[-n]/|h[-n]|$, then $|x[n]| \equiv 1$ bounded

$$\text{But } y[0] = \sum_{k=-\infty}^{\infty} x[k]h[-k] = \sum_{k=-\infty}^{\infty} h^*[-k]h[-k]/|h[-k]| = \sum_{k=-\infty}^{\infty} |h[-k]| = \infty$$

- Memoryless / with Memory

- A linear, time-invariant, causal system is memoryless only

if $h[n] = K\delta[n]$ $h(t) = K\delta(t)$

$y[n] = Kx[n]$ $y(t) = Kx(t)$

if $K=1$ further, they are identity systems

$$y[n] = x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k] = x[n] * \delta[n]$$

$$y(t) = x(t) = \int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)d\tau = x(t) * \delta(t)$$

Summary

Understand the following new concepts:

- 1. Use unit impulse function to represent any function**
- 2. Unit impulse response $h[n]$**
 - ◆ Given the system input/output equation, how to decide the unit impulse response?
- 3. Convolution, its properties, and calculation steps (FSMS)**
 - ◆ Understand the meaning of index 'k' and index 'n'
- 4. Decide LTI system property by using unit impulse response $h[n]$**